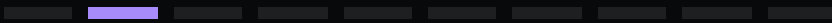


**Verification cost
scales with the
number of distinct
mechanisms.**

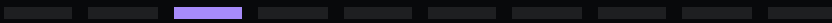
measured by kinds, not by component count



THE NAIVE COUNT

**Four hundred
components, four
hundred verification
problems.**

Round numbers, used to show how the cost behaves. Every component that acts on the world needs somebody to establish that it behaves under load, under failure, and under an input nobody anticipated.



WHAT IS ACTUALLY THERE

**Four hundred
components
are rarely four
hundred kinds.**

They are a smaller set of kinds,
instantiated repeatedly with
different parameters.



WHERE VERIFICATION ATTACHES

Verification attaches to the kind.

Prove the kind once, with a
theorem or a harness, and every
instance inherits the result.

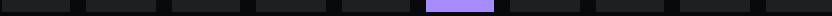


THE CURVE

400 → 15

Illustrative figures. Cost stops tracking how much system there is and starts tracking how many distinct mechanisms it contains.

not a measurement



THE CONDITION

**The instances have
to be genuinely
instances.**

One component patched
differently from its siblings stops
inheriting the proof and becomes
its own kind, priced separately.



WHY THE COUNT IS WRONG

Systems accumulate special cases.

The count of them is rarely maintained anywhere, which means the verification bill is routinely estimated from the wrong number.



HOW TO PRICE THE DIFFERENCE

In the scarcest input.

Disciplines that have to sit in one team simultaneously, and operating time that cannot be parallelised. Currency is the input a funded buyer can always obtain.

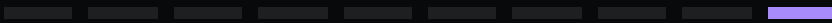


WHERE THIS STOPS

**It assumes the set
of kinds is stable.**

While the mechanism set is still
moving, the target moves with it
and the curve does not yet apply.

the argument has an edge, and this is it



THE TEST

How many kinds does your system contain?

The count of things that are actually different, which is set by design decisions rather than discovered by an audit.